

COMP0005 Algorithms: Group Coursework Report

Group Number: XXX

Academic year 2025/26

1 Experimental framework overview

(Section 1 should be about one page in length.)

1.1 Framework design and architecture

The framework empirically measures the asymptotic relationship between the operation (search, insertion) times and number of nodes in a balanced tree.

Synthetic keys are generated in the *TestDataGenerator*, which contains four generators to suitably stress test the trees: sequential (ascending, descending), and random (uniform, normal). Uniform is the best case, as keys are naturally self balancing, (expected height of a vanilla BST is $O(\log n)$ [1]). Normal is the average case, as due to Central Limit Theorem, natural data often takes a normal distribution. Sequential are the worst cases, where maximal rebalances occur due to always inserting on the left or right of the tree.

The time to compare keys is a control variable, so the framework ensures all keys are a constant string length, k . This makes the time to compare keys $O(k)$. As k is constant, the cost of key comparisons is $O(1)$ with respect to the number of nodes, n , in the tree. This ensures the theoretical time complexity of operation is still $O(\log n)$. As keys are generated numerically and cast to strings, the intended random distributions change due to the difference in numerical and lexicographic order. To maintain numerical ordering, truncating (casting to Python *int*), casting to *str*, and left padding with 0's is used. The string length of 7 was chosen as it allows for 10^7 unique keys, more than the number of nodes being inserted.

Data is collected using the *ExperimentalFramework*. The algorithm builds m trees up to the maximum size n while measuring operation cost at intervals defined on a geometric progression, ratio $r = e^{\ln(n)/data\ points}$. The geometric progression spreads the data points equally when plotting a log-graph. To minimize bias from OS processes and cache effects, m is sufficiently large. To measure the asymptotic relationship, n is sufficiently large. To be confident in our data, r is sufficiently low. At each interval, an insertion is measured along with a search for a key inside and outside the tree.

After collecting the data, it is processed using the *StatisticsFramework*. Log values of n at each interval are calculated as this transforms the log relationship to a linear one. The framework then calculates the PMCC, a quantitative measurement of linearity, to verify the relationship. The regression line is calculated as it can be used to reason about the hidden constants of different tree implementations. The RMSE is calculated as it allows for

reasoning about the predictability of a tree implementation's performance. The residuals are plotted as they can reveal hidden mechanics about the trees (explored in section 2.1).

Finally, graphs are processed using Matplotlib.

1.2 Hardware/software setup for experiments

Hardware:

- Laptop: Apple MacBook Pro M4 (10 core CPU, 10 core GPU)
- Memory: 16GB LPDDR5X; 120 Gbs⁻¹ bandwidth [2]
- Cache (`$ sysctl -a | grep perflevel`):
 - L1: 192 (instruction) + 128 KiB (data) per P core; 128 + 64 KiB per E core
 - L2: 16 MiB for P core cluster; 4 MiB for E core cluster

Software:

- OS: MacOS Tahoe 26.3.1
- Python Interpreter: 3.14
- Python Libraries: timeit, random, matplotlib, math

2 Performance results

(Section 2 should be about two to two-and-a-half pages in length.)

2.1 2-3 Tree

2.2 LLRB BST

2.3 AVL Tree

3 Comparative assessment

(Section 3 should be about one to one-and-a-half pages in length.)

4 Team contributions

(Section 4 should be no more than half a page in length. Express work share as a percentage totalling to 100: for example, four students who all contributed equally should receive 25%.)

Name	Portico ID	Key contributions	Work share (%)
(name)	(number)	(text)	(xx%)

References

- [1] T. H. Cormen, *Introduction to Algorithms 3rd ed.* MIT Press, 2009.
- [2] Wikipedia, *Apple m4*, https://en.wikipedia.org/wiki/Apple_M4, 24 Mar 2026, Accessed 27 Mar 2026.